RealSystem Pvt. Ltd.

Crude Oil Transfer Pump Station

Safety Control Narrative - Unit RS-100

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Applicable Standard:	IEC 61511-1:2016 / ISA-84.00.01

CAUTION: This document contains safety-critical control logic for the RS-100 crude oil transfer pump station. All shutdown setpoints and interlock parameters must be verified by a qualified process safety engineer before commissioning. Unauthorized modification of logic described herein may result in loss of containment, fire, explosion, or fatality.

1. Introduction and Scope

This Control Narrative defines the automated safety and control logic for the RealSystem RS-100 Crude Oil Transfer Pump Station located at the Kakinada Terminal Facility. The pump station transfers crude oil from the tank farm (TK-001 to TK-004) to the pipeline manifold at a nominal flow rate of 250 m3/h at a discharge pressure of 45 to 60 bar.

This document covers:

- Normal operating procedures and start/stop sequences
- Alarm setpoints and operator response guidance
- Safety instrumented system (SIS) shutdown logic
- Emergency shutdown (ESD) philosophy
- Interlock descriptions and IEC 61511 invariant mapping

All safety logic described in this document is implemented in the RealSystem Safety PLC (model RS-SIL2-PLC) and verified to SIL 2 per IEC 61511.

2. Equipment Register

The following equipment is addressed by this control narrative:

Tag	Type	Description	Process Connection
P-001	Centrifugal Pump	Crude Oil Transfer Pump #1 (Duty)	Suction: 16" ANSI 300, Discharge: 12" ANSI 60
P-002	Centrifugal Pump	Crude Oil Transfer Pump #2 (Standby)	Suction: 16" ANSI 300, Discharge: 12" ANSI 60
PT-001	Pressure Transmitter	P-001/P-002 Common Discharge Pressure	1" NPT on 12" discharge header
PT-002	Pressure Transmitter	Pump Suction Header Pressure	1" NPT on 16" suction header
FT-001	Flow Transmitter	Discharge Pipeline Flow (Coriolis)	4" inline on discharge header
LT-001	Level Transmitter	Feed Tank TK-001 Level (Guided Radar)	Tank nozzle N7, top of TK-001
TT-001	Temperature Transmitter	P-001 Drive-End Bearing Temperature	PT100 in bearing housing
TT-002	Temperature Transmitter	P-002 Drive-End Bearing Temperature	PT100 in bearing housing
XV-001	Motorised Ball Valve	P-001 Discharge Isolation Valve	12" ANSI 600, fail-closed
XV-002	Motorised Ball Valve	P-002 Discharge Isolation Valve	12" ANSI 600, fail-closed
XV-003	Motorised Ball Valve	Suction Header Isolation Valve	16" ANSI 300, fail-closed
ESD-001	Emergency Shutdown	Station-Wide Emergency Shutdown System	Hardwired to SIL 2 Safety PLC
I-001	Safety Interlock	High-High Discharge Pressure Interlock	Trips P-001, P-002 on PT-001 HH
I-002	Safety Interlock	Low-Low Suction Pressure Interlock	Trips P-001, P-002 on PT-002 LL
I-003	Safety Interlock	Low-Low Tank Level Interlock	Trips P-001, P-002 on LT-001 LL
I-004	Safety Interlock	High Bearing Temperature Interlock	Trips affected pump on TT-001/TT-002 HH

3. Normal Operating Conditions

3.1 Pump Start Sequence

Prior to starting P-001 or P-002, the following permissives must be satisfied:

(a) XV-003 (suction isolation valve) is confirmed OPEN

(b) PT-002 (suction pressure) reads >= 3.5 bar

(c) LT-001 (tank level) reads >= 30%

(d) No active HIGH or HIGH-HIGH alarms on TT-001 (for P-001) or TT-002 (for P-002)

(e) ESD-001 is in RESET state (not tripped)

The start sequence is: (1) Open discharge isolation valve XV-001 (or XV-002 for P-002), (2) Start pump motor, (3) Confirm flow indication on FT-001 > 30 m3/h within 15 seconds. If flow is not confirmed, the pump shall trip and a START FAILURE alarm shall be raised.

3.2 Normal Operating Ranges

Under steady-state operation the following process parameters are expected:

Parameter	Tag	Normal Range	Units
Discharge Pressure	PT-001	45 - 60	bar
Suction Pressure	PT-002	3.5 - 6.0	bar
Pipeline Flow	FT-001	200 - 280	m3/h
Tank TK-001 Level	LT-001	30 - 90	% of height
P-001 Bearing Temperature	TT-001	40 - 65	degC
P-002 Bearing Temperature	TT-002	40 - 65	degC

3.3 Duty/Standby Changeover

P-001 is designated as the duty pump and P-002 as standby. Automatic changeover from P-001 to P-002 occurs if:

- P-001 trips on any interlock condition, OR
- P-001 motor fault signal is received from the MCC

Manual changeover is permitted from the DCS only when both pumps are in READY state and no alarms are active. Changeover logic ensures XV-002 is fully open before P-002 starts, preventing dead-head startup.

4. Alarm and Trip Setpoints

The following table defines all alarm and trip setpoints for the RS-100 pump station. HIGH (H) alarms are operator warnings. HIGH-HIGH (HH) and LOW-LOW (LL) setpoints trigger automatic safety actions via the SIS interlocks described in Section 5.

Tag	Alarm Level	Setpoint	Unit	Action	IEC Clause
PT-001	HIGH (H)	65	bar	ALARM - operator to reduce flow or check for	IEC 61511 Cl.9.2.2
PT-001	HIGH-HIGH (HH	75	bar	TRIP P-001 and P-002 via I-001. Close XV-001	IEC 61511 Cl.9.2.2
PT-002	LOW (L)	3.5	bar	ALARM - operator to check suction valve and	IEC 61511 Cl.9.2.2
PT-002	LOW-LOW (LL)	2.0	bar	TRIP P-001 and P-002 via I-002. Close XV-003	IEC 61511 Cl.9.2.2
FT-001	LOW (L)	50	m3/h	ALARM - check for valve restriction or pump c	IEC 61511 Cl.9.2.3
FT-001	LOW-LOW (LL)	20	m3/h	ALARM - deadhead condition suspected, oper	IEC 61511 Cl.9.2.3
LT-001	LOW (L)	25	% ht	ALARM - switch to alternate tank or reduce flo	IEC 61511 Cl.9.2.2
LT-001	LOW-LOW (LL)	10	% ht	TRIP P-001 and P-002 via I-003. Prevent cavita	IEC 61511 Cl.9.2.2
TT-001	HIGH (H)	75	degC	ALARM - P-001 bearing temperature rising	IEC 61511 Cl.9.2.3
TT-001	HIGH-HIGH (HH	90	degC	TRIP P-001 via I-004. Protect pump bearings	IEC 61511 Cl.9.2.3
TT-002	HIGH (H)	75	degC	ALARM - P-002 bearing temperature rising	IEC 61511 Cl.9.2.3
TT-002	HIGH-HIGH (HH	90	degC	TRIP P-002 via I-004. Protect pump bearings	IEC 61511 Cl.9.2.3

5. Safety Interlock and Shutdown Logic

This section describes all safety interlocks implemented in the RealSystem SIL 2 Safety PLC. Each interlock is associated with an IEC 61511 invariant requirement. The interlock logic uses 1oo1 voting (single transmitter) for most functions; PT-001 HH shutdown uses 2oo3 voting per the SIL 2 requirement.

5.1 Interlock I-001 - High-High Discharge Pressure Shutdown

Sensor: PT-001 (Discharge Pressure Transmitter)
Trip Level: HIGH-HIGH at 75 bar
Voting: 2oo3 (three independent pressure transmitters PT-001A, PT-001B, PT-001C)
SIL Target: SIL 2 (IEC 61511-1 Clause 9.2.2)

When discharge pressure sensor PT-001 reads greater than or equal to 75 bar, interlock I-001 shall activate immediately. Upon activation of I-001, pump P-001 shall be stopped and pump P-002 shall be stopped. Additionally, discharge isolation valves XV-001 and XV-002 shall be closed. The interlock shall latch and require manual reset from the DCS by an authorised operator after the root cause has been identified and resolved.

Pre-trip alarm: PT-001 >= 65 bar raises a HIGH pressure alarm to give the operator a 30-second window to take corrective action before the automatic trip fires.

5.2 Interlock I-002 - Low-Low Suction Pressure Shutdown

Sensor: PT-002 (Suction Pressure Transmitter)
Trip Level: LOW-LOW at 2.0 bar
Voting: 1oo1
SIL Target: SIL 1 (IEC 61511-1 Clause 9.2.2)

When suction pressure sensor PT-002 reads less than or equal to 2.0 bar, interlock I-002 shall activate. Upon activation, pump P-001 shall be stopped and pump P-002 shall be stopped to prevent cavitation and mechanical damage to pump seals. Suction isolation valve XV-003 shall be closed to isolate the suction header. The interlock shall latch pending manual reset.

5.3 Interlock I-003 - Low-Low Feed Tank Level Shutdown

Sensor: LT-001 (Tank TK-001 Level Transmitter)
Trip Level: LOW-LOW at 10% of tank height
Voting: 1oo1
SIL Target: SIL 1 (IEC 61511-1 Clause 9.2.2)

When feed tank level transmitter LT-001 reads less than or equal to 10 percent of tank height, interlock I-003 shall activate. Upon activation, pump P-001 shall be stopped and pump P-002 shall be stopped to prevent dry running and pump cavitation. A LOW-LOW LEVEL alarm shall be raised on the DCS. The interlock shall latch pending manual reset after tank level is restored above 20 percent.

5.4 Interlock I-004 - High Bearing Temperature Shutdown

Sensor: TT-001 (P-001 Bearing Temperature) / TT-002 (P-002 Bearing Temperature)
Trip Level: HIGH-HIGH at 90 degC
Voting: 1oo1 per pump (independent for each pump)
SIL Target: SIL 1 (IEC 61511-1 Clause 9.2.3)

When bearing temperature transmitter TT-001 reads greater than or equal to 90 degC, interlock I-004 shall trip pump P-001 only. P-001 shall be stopped immediately and its discharge isolation valve XV-001 shall close. P-002 may continue or start as standby pump.

Similarly, when bearing temperature transmitter TT-002 reads greater than or equal to 90 degC, interlock I-004 shall trip pump P-002 only. XV-002 shall close. P-001 may continue or start.

Pre-trip alarm at TT-001 >= 75 degC and TT-002 >= 75 degC provides early warning.

6. Emergency Shutdown (ESD) Philosophy

The Emergency Shutdown System ESD-001 is a station-wide safety layer independent of the process control system. ESD-001 activates under any of the following conditions:

(a) Manual pushbutton activation from the local field panel or control room
(b) Loss of instrument air supply (< 4.5 bar)
(c) Loss of 24VDC safety bus
(d) Fire and Gas system F&G-001 trip signal received
(e) High-High pressure on PT-001 confirmed by 2oo3 voting (redundant path to I-001)

Upon activation of ESD-001, the following actions are executed simultaneously and unconditionally within 500 milliseconds:

1. Pump P-001 is stopped
2. Pump P-002 is stopped
3. Discharge isolation valve XV-001 is closed (fail-closed actuator)
4. Discharge isolation valve XV-002 is closed (fail-closed actuator)
5. Suction isolation valve XV-003 is closed (fail-closed actuator)
6. ESD active signal is sent to the pipeline SCADA system

ESD-001 is a hardwired latch circuit. Reset requires physical key-switch at the local panel and DCS confirmation from a Level 3 authorised engineer. Bypassing ESD-001 is not permitted during normal operation and requires a Management of Change (MoC) approval.

7. Cause and Effect Summary

The following cause-and-effect matrix summarises all shutdown actions:

Cause (Initiating Event)	Tag	Setpoint	Effect: P-001	Effect: P-002	Effect: Valves
Discharge pressure HIGH-HIGH	PT-001	>= 75 bar	STOP	STOP	CLOSE XV-001, XV-002
Suction pressure LOW-LOW	PT-002	<= 2.0 bar	STOP	STOP	CLOSE XV-003
Tank level LOW-LOW	LT-001	<= 10% ht	STOP	STOP	No valve action
P-001 bearing temp HIGH-HIGH	TT-001	>= 90 degC	STOP	RUN	CLOSE XV-001 only
P-002 bearing temp HIGH-HIGH	TT-002	>= 90 degC	RUN	STOP	CLOSE XV-002 only
ESD-001 activated	ESD-001	Any trigger	STOP	STOP	CLOSE XV-001, XV-002, XV-003

8. Control Rule Summary (Machine-Readable Format)

The following rules are extracted directly from Sections 4 and 5 for use in the RealSystem Automated Safety Audit System (RS-ASAS). Each rule is stated in IF/THEN form with explicit sensor tags, numeric thresholds, and equipment actions.

R-001	IF PT-001 >= 75 bar	STOP P-001	IEC 61511 Cl.9.2.2
R-002	IF PT-001 >= 75 bar	STOP P-002	IEC 61511 Cl.9.2.2
R-003	IF PT-001 >= 65 bar	ALARM P-001	IEC 61511 Cl.9.2.2
R-004	IF PT-002 <= 2.0 bar	STOP P-001	IEC 61511 Cl.9.2.2
R-005	IF PT-002 <= 2.0 bar	STOP P-002	IEC 61511 Cl.9.2.2
R-006	IF PT-002 <= 3.5 bar	ALARM P-001	IEC 61511 Cl.9.2.2
R-007	IF LT-001 <= 10%	STOP P-001	IEC 61511 Cl.9.2.2
R-008	IF LT-001 <= 10%	STOP P-002	IEC 61511 Cl.9.2.2
R-009	IF LT-001 <= 25%	ALARM P-001	IEC 61511 Cl.9.2.2
R-010	IF TT-001 >= 90 degC	STOP P-001	IEC 61511 Cl.9.2.3
R-011	IF TT-002 >= 90 degC	STOP P-002	IEC 61511 Cl.9.2.3
R-012	IF TT-001 >= 75 degC	ALARM P-001	IEC 61511 Cl.9.2.3
R-013	IF TT-002 >= 75 degC	ALARM P-002	IEC 61511 Cl.9.2.3
R-014	IF FT-001 <= 50 m3/h	ALARM P-001	IEC 61511 Cl.9.2.3
R-015	IF ESD-001 == ACTIVE	STOP P-001	IEC 61511 Cl.9.2.2
R-016	IF ESD-001 == ACTIVE	STOP P-002	IEC 61511 Cl.9.2.2
R-017	IF ESD-001 == ACTIVE	CLOSE XV-001	IEC 61511 Cl.9.2.2

R-018	IF ESD-001 == ACTIVE	CLOSE XV-002	IEC 61511 Cl.9.2.2

R-019	IF ESD-001 == ACTIVE	CLOSE XV-003	IEC 61511 Cl.9.2.2

9. Document Revision History

Rev	Date	Author	Description of Change
1.0	2023-03-10	P. Safety Group	Initial issue for HAZOP review
1.1	2023-06-22	P. Safety Group	Incorporated HAZOP action items A-014 to A-031
2.0	2024-01-15	Lead ICS Safety Engineer	Added TT-001/TT-002 bearing temp interlocks (I-004)
2.1	2024-09-15	P. Safety Group	Updated PT-001 trip setpoint from 70 to 75 bar (MoC-2024-017)

10. Applicable Standards and References

1. IEC 61511-1:2016 - Functional safety: Safety instrumented systems for the process industry sector - Part 1: Framework, definitions, system, hardware and application programming requirements.

2. ISA-84.00.01-2004 (IEC 61511 Mod) - Functional Safety: Safety Instrumented Systems for the Process Industry Sector.

3. IEC 61508-1:2010 - Functional safety of E/E/PE safety-related systems.

4. API RP 14C - Analysis, Design, Installation, and Testing of Basic Surface Safety Systems for Offshore Production Platforms.

5. RealSystem Internal Standard RS-ENG-005 Rev 3 - Safety PLC Programming Guidelines.

6. RealSystem SIL Determination Study - DOC-RS-SIL-001 Rev 1.0.

7. RealSystem Piping and Instrumentation Diagram - DWG-RS-PID-100-001 Rev 4.

END OF DOCUMENT - DOC-RS-CN-001 Rev 2.1

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